$$C_{qd}^{(1)}\,\text{i1\_i2\_i3\_i4\_} \to$$

$$\frac{1}{16\,\pi^2}\,\Bigg(-\frac{2}{243}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{324}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{1215}\sum_p g_1^4$$

$$LF_{5,-2}\left[m_{\tilde{d}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{108}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{8}{405}\sum_p g_1^4\,LF_{5,-2}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{216}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{405}\sum_p g_1^4\,LF_{5,-2}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{243}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{648}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{1215}\sum_p g_1^4\,LF_{5,-2}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{243}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{81}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{1215}\sum_p g_1^4\,LF_{5,-2}\left[m_{\tilde{u}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^4\,LF_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}\,g_1^4\,LF_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{405}\,g_1^4\,LF_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,LF_{2,1,0}\left[m_1,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,LF_{2,2,-1}\left[m_1,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}\,g_1^4\,LF_{3,1,-1}\left[m_1,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,LF_{4,1,-2}\left[m_1,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,LF_{2,1,0}\left[m_1,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,LF_{2,2,-1}\left[m_1,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}\,g_1^4\,LF_{3,1,-1}\left[m_1,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,LF_{4,1,-2}\left[m_1,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}\,g_1^4\,LF_{2,1,0}\left[m_1,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{2,2,-1}\left[m_1,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,LF_{3,1,-1}\left[m_1,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{4,1,-2}\left[m_1,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}\,g_1^4\,LF_{2,1,0}\left[m_1,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{2,2,-1}\left[m_1,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,LF_{3,1,-1}\left[m_1,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{4,1,-2}\left[m_1,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}\,g_1^2\,g_2^2\,LF_{2,1,0}\left[m_2,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{2,2,-1}\left[m_2,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,LF_{3,1,-1}\left[m_2,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{4,1,-2}\left[m_2,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}\,g_1^2\,g_2^2\,LF_{2,1,0}\left[m_2,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{2,2,-1}\left[m_2,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,LF_{3,1,-1}\left[m_2,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{4,1,-2}\left[m_2,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_3,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,2,-1}\left[m_3,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_3,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{4,1,-2}\left[m_3,\,m_{\tilde{d}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_3,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,2,-1}\left[m_3,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_3,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{4,1,-2}\left[m_3,\,m_{\tilde{d}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_3,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,2,-1}\left[m_3,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_3,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{4,1,-2}\left[m_3,\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_3,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{2,2,-1}\left[m_3,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_3,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{4,1,-2}\left[m_3,\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,1,0}\left[m_{\tilde{d}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}+\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,2,-1}\left[m_{\tilde{d}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{3,1,-1}\left[m_{\tilde{d}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}+\frac{1}{486}\,g_1^4\,LF_{2,1,0}\left[m_{\tilde{d}}^{i3},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,LF_{3,1,-1}\left[m_{\tilde{d}}^{i3},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_{\tilde{d}}^{i3},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_{\tilde{d}}^{i3},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{486}\,g_1^4\,LF_{2,1,0}\left[m_{\tilde{d}}^{i4},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,LF_{3,1,-1}\left[m_{\tilde{d}}^{i4},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_{\tilde{d}}^{i4},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_{\tilde{d}}^{i4},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{6}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{3,1,0}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]+$$

$$\frac{1}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{4,1,-1}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]-\frac{1}{3}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{5,1,-2}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]-$$

$$\frac{1}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,1,0}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]+\frac{3}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{4,1,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]-$$

$$\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{5,1,-2}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]-\frac{1}{54}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,1,0}\left[m_{\tilde{q}}^{p},\,\tilde{\mu}\right]\delta_{i1i2}+$$

$$\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,2,-1}\left[m_{\tilde{q}}^{p},\,\tilde{\mu}\right]\delta_{i1i2}+\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{3,1,-1}\left[m_{\tilde{q}}^{p},\,\tilde{\mu}\right]\delta_{i1i2}+$$

$$\frac{1}{1944}\,g_1^4\,LF_{2,1,0}\left[m_{\tilde{q}}^{i1},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i1},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,LF_{2,1,0}\left[m_{\tilde{q}}^{i1},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i1},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_{\tilde{q}}^{i1},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i1},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,LF_{2,1,0}\left[m_{\tilde{q}}^{i2},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i2},\,m_1\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,LF_{2,1,0}\left[m_{\tilde{q}}^{i2},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i2},\,m_2\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,LF_{2,1,0}\left[m_{\tilde{q}}^{i2},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,LF_{3,1,-1}\left[m_{\tilde{q}}^{i2},\,m_3\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,1,0}\left[m_{\tilde{u}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}-\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,2,-1}\left[m_{\tilde{u}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{3,1,-1}\left[m_{\tilde{u}}^{p},\,\tilde{\mu}\right]\delta_{i3i4}-\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,1,0}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]+$$

$$\frac{3}{2}\,\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{4,1,-1}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]-\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{5,1,-2}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]+$$

$$\frac{1}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{3,1,-1}[\tilde{\mu},\,m_1]-\frac{2}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{4,1,-2}[\tilde{\mu},\,m_1]+$$

$$\frac{1}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{5,1,-3}[\tilde{\mu},\,m_1]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{3,1,-1}[\tilde{\mu},\,m_2]-$$

$$2\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{4,1,-2}[\tilde{\mu},\,m_2]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{5,1,-3}[\tilde{\mu},\,m_2]+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,1,0}\left[\tilde{\mu},\,m_{\tilde{d}}^{p}\right]\delta_{i3i4}+\frac{5}{108}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{3,1,-1}\left[\tilde{\mu},\,m_{\tilde{d}}^{p}\right]\delta_{i3i4}-$$

$$\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{4,1,-2}\left[\tilde{\mu},\,m_{\tilde{d}}^{p}\right]\delta_{i3i4}+\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,1,0}\left[\tilde{\mu},\,m_{\tilde{q}}^{p}\right]\delta_{i1i2}+$$

$$\frac{7}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{3,1,-1}\left[\tilde{\mu},\,m_{\tilde{q}}^{p}\right]\delta_{i1i2}-\frac{1}{54}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{4,1,-2}\left[\tilde{\mu},\,m_{\tilde{q}}^{p}\right]\delta_{i1i2}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,1,0}\left[\tilde{\mu},\,m_{\tilde{u}}^{p}\right]\delta_{i3i4}-\frac{1}{108}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{3,1,-1}\left[\tilde{\mu},\,m_{\tilde{u}}^{p}\right]\delta_{i3i4}-$$

$$\frac{1}{108}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{4,1,-2}\left[\tilde{\mu},\,m_{\tilde{u}}^{p}\right]\delta_{i3i4}-\frac{1}{648}\,g_1^4\,LF_{2,1,1,-1}\left[m_1,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{324}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\left[m_1,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{216}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,2,1,-1}\left[m_1,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i2}\right]-\frac{1}{216}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}\left[m_1,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i1}\right]-$$

$$\frac{1}{648}\,g_1^4\,LF_{2,1,1,-1}\left[m_1,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{324}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\left[m_1,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{216}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}\left[m_1,\,m_{\tilde{q}}^{i1},\,m_{\tilde{d}}^{i4}\right]-$$

$$\frac{1}{216}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}\left[m_1,\,m_{\tilde{q}}^{i2},\,m_{\tilde{d}}^{i3}\right]-\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_1,\,\tilde{\mu},\,m_{\tilde{d}}^{i3}\right]-$$

$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_1,\,\tilde{\mu},\,m_{\tilde{d}}^{i4}\right]-\frac{1}{72}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_1,\,\tilde{\mu},\,m_{\tilde{q}}^{i1}\right]-$$

$$\frac{1}{72}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_1,\,\tilde{\mu},\,m_{\tilde{q}}^{i2}\right]-\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_2,\,\tilde{\mu},\,m_{\tilde{q}}^{i1}\right]-$$

$$\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}\left[m_2,\,\tilde{\mu},\,m_{\tilde{q}}^{i2}\right]-\frac{1}{9}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}\left[m_3,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i2}\right]+$$

$$\frac{1}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}\left[m_3,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i1}\right]-\frac{1}{9}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}\left[m_3,\,m_{\tilde{q}}^{i1},\,m_{\tilde{d}}^{i4}\right]+$$

$$\frac{1}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}\left[m_3,\,m_{\tilde{q}}^{i2},\,m_{\tilde{d}}^{i3}\right]+\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,1,1,0}\left[m_{\tilde{d}}^{i3},\,m_1,\,m_{\tilde{q}}^{i2}\right]-\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}\left[m_{\tilde{d}}^{i3},\,m_1,\,m_{\tilde{q}}^{i2}\right]-$$

$$\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,1,1,0}\left[m_{\tilde{d}}^{i3},\,m_3,\,m_{\tilde{q}}^{i2}\right]+\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{3,1,1,-1}\left[m_{\tilde{d}}^{i3},\,m_3,\,m_{\tilde{q}}^{i2}\right]+\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}\left[m_{\tilde{d}}^{i4},\,m_1,\,m_{\tilde{q}}^{i1}\right]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}\left[m_{\tilde{d}}^{i4},\,m_1,\,m_{\tilde{q}}^{i1}\right]-\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$LF_{2,1,1,0}\left[m_{\tilde{d}}^{i4},\,m_3,\,m_{\tilde{q}}^{i1}\right]+\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}\left[m_{\tilde{d}}^{i4},\,m_3,\,m_{\tilde{q}}^{i1}\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{2,1,1,0}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r},\,\tilde{\mu}\right]+$$

$$\frac{1}{12}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{3,1,1,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r},\,\tilde{\mu}\right]+$$

$$\frac{1}{24}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{2,2,1,-1}\left[m_{\tilde{q}}^{p},\,\tilde{\mu},\,m_{\tilde{u}}^{r}\right]+$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}\left[m_{\tilde{q}}^{i1},\,m_1,\,m_{\tilde{d}}^{i4}\right]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}\left[m_{\tilde{q}}^{i1},\,m_1,\,m_{\tilde{d}}^{i4}\right]-$$

$$\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}\left[m_{\tilde{q}}^{i1},\,m_3,\,m_{\tilde{d}}^{i4}\right]+\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$LF_{3,1,1,-1}\left[m_{\tilde{q}}^{i1},\,m_3,\,m_{\tilde{d}}^{i4}\right]+\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,1,1,0}\left[m_{\tilde{q}}^{i2},\,m_1,\,m_{\tilde{d}}^{i3}\right]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}\left[m_{\tilde{q}}^{i2},\,m_1,\,m_{\tilde{d}}^{i3}\right]-\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,1,1,0}\left[m_{\tilde{q}}^{i2},\,m_3,\,m_{\tilde{d}}^{i3}\right]+\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}\left[m_{\tilde{q}}^{i2},\,m_3,\,m_{\tilde{d}}^{i3}\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{2,1,1,0}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p},\,\tilde{\mu}\right]+$$

$$\frac{1}{12}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{3,1,1,-1}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p},\,\tilde{\mu}\right]+$$

$$\frac{1}{24}\,\tilde{\mu}^2\left(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\right)LF_{2,2,1,-1}\left[m_{\tilde{u}}^{r},\,\tilde{\mu},\,m_{\tilde{q}}^{p}\right]-$$

$$\frac{1}{4}\,\overline{y_d}^{ri3}\,\overline{y_d}^{i2p}\,y_d^{ri4}\,y_d^{i1p}\,LF_{2,1,1,-1}\left[\tilde{\mu},\,m_{\tilde{d}}^{p},\,m_{\tilde{q}}^{r}\right]-$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\overline{y_u}^{i2r}\,y_u^{i1r}\,LF_{2,1,1,0}\left[\tilde{\mu},\,m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r}\right]+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_1,\,m_{\tilde{d}}^{i3},\,m_{\tilde{d}}^{i4},\,\tilde{\mu}\right]+\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}\left[m_1,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1},\,\tilde{\mu}\right]+\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_1,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2},\,\tilde{\mu}\right]+$$

$$\frac{1}{216}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_1,\,m_{\tilde{q}}^{i1},\,m_{\tilde{q}}^{i2},\,\tilde{\mu}\right]+\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}\left[m_2,\,m_{\tilde{q}}^{i1},\,m_{\tilde{q}}^{i2},\,\tilde{\mu}\right]+\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_3,\,m_{\tilde{d}}^{i3},\,m_{\tilde{d}}^{i4},\,\tilde{\mu}\right]-$$

$$\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_3,\,m_{\tilde{d}}^{i3},\,m_{\tilde{q}}^{i1},\,\tilde{\mu}\right]-\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}\left[m_3,\,m_{\tilde{d}}^{i4},\,m_{\tilde{q}}^{i2},\,\tilde{\mu}\right]+\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}\left[m_3,\,m_{\tilde{q}}^{i1},\,m_{\tilde{q}}^{i2},\,\tilde{\mu}\right]\Bigg)$$